

Sublinear Kalton–Peck centralizers are not singular

Candidate negative answer to the question in arXiv:1410.5505

19 July 2026

Status: candidate full negative answer; human review required. For every $0 < r < 1$, the Kalton–Peck centralizer $\mathcal{K}_{\phi_r}$ on ℓ_2 is not singular. The proof exhibits a disjointly generated copy of ℓ_2 on which $\mathcal{K}_{\phi_r}$ is trivial.

1 Source and exact question

The source is J. M. F. Castillo, V. Ferenczi, and M. González, *Singular twisted sums generated by complex interpolation*, arXiv:1410.5505v1 (2014) [1]. At the end of Section 6, page 25 of the arXiv PDF, the authors ask whether the sublinear Kalton–Peck maps are singular:

Recall from [32, Corollary 5.5] that the spaces $\ell_2(\phi_r)$ are mutually non-isomorphic for different values of $0 < r \leq 1$. We already know [32, Corollary 5.5] that $\mathcal{K} = \mathcal{K}_{\phi_1}$ is singular but, since the function ϕ_r is not expansive for $r < 1$, we do not know if also $\mathcal{K}_{\phi_r}$ is singular for $0 < r < 1$.

Figure 1: The open question rendered from page 25 of the source PDF.

Transcription. “We already know [32, Corollary 5.5] that $\mathcal{K} = \mathcal{K}_{\phi_1}$ is singular but, since the function ϕ_r is not expansive for $r < 1$, we do not know if also $\mathcal{K}_{\phi_r}$ is singular for $0 < r < 1$.”

2 Definitions

For $0 < r \leq 1$, set

$$\phi_r(t) = \begin{cases} t, & 0 \leq t \leq 1, \\ t^r, & t > 1. \end{cases}$$

The associated Kalton–Peck centralizer on ℓ_2 is

$$\mathcal{K}_{\phi_r}(x) = x \phi_r \left(-\log \frac{|x|}{\|x\|_2} \right),$$

with all operations taken coordinatewise. A quasi-linear map is *trivial on a subspace* Y if there is a linear map $L : Y \rightarrow \mathbb{C}^{\mathbb{N}}$ such that $\mathcal{K}_{\phi_r} - L : Y \rightarrow \ell_2$ is bounded. It is singular exactly when its restriction to every infinite-dimensional closed subspace is nontrivial; equivalently, the quotient map in the associated twisted sum is strictly singular [1, Section 4].

3 Proof Intuition

On a normalized flat block of length m , every nonzero coordinate has logarithmic size $s = \frac{1}{2} \log m$. A linear diagonal map can remove the base value s^r from the centralizer. For a linear combination with coefficient a , the remaining nonlinear error is

$$(s - \log |a|)^r - s^r.$$

Because $r < 1$, increments of t^r flatten as the base point s grows. We choose successive block sizes so rapidly increasing that these increments are square summable uniformly over every unit vector. The only apparent danger is from extremely small coefficients, but their exponential weight $|a|^2 = e^{-2t}$ beats the power t^{2r} . This produces an explicit infinite-dimensional subspace on which the centralizer is trivial.

4 Negative answer

Theorem 1. *For every $0 < r < 1$, the centralizer $\mathcal{K}_{\phi_r}$ on ℓ_2 is not singular. More precisely, there are pairwise disjoint finite sets $A_n \subset \mathbb{N}$ such that, for*

$$u_n = |A_n|^{-1/2} \mathbf{1}_{A_n}, \quad Y = \overline{\text{span}}\{u_n : n \geq 1\},$$

the restriction $\mathcal{K}_{\phi_r}|_Y$ is trivial. Consequently, the quotient map of $\ell_2(\phi_r) = \ell_2 \oplus_{\mathcal{K}_{\phi_r}} \ell_2$ is not strictly singular.

Proof. Fix $0 < r < 1$ and put

$$\beta = 2(1 - r) > 0.$$

Choose $a > 1/(2\beta) = 1/(4(1 - r))$. Take pairwise disjoint finite sets A_n with cardinalities m_n so large that

$$s_n := \frac{1}{2} \log m_n \geq \max\{1, n^a\}.$$

The disjoint normalized vectors $u_n = m_n^{-1/2} \mathbf{1}_{A_n}$ are isometric to the unit vector basis of ℓ_2 , so $Y \cong \ell_2$.

Define a linear, coordinatewise map $L : Y \rightarrow \mathbb{C}^{\mathbb{N}}$ by

$$L(u_n) = s_n^r u_n.$$

We prove that $\mathcal{K}_{\phi_r} - L$ is bounded from Y to ℓ_2 . By homogeneity it suffices to consider

$$x = \sum_n a_n u_n, \quad \sum_n |a_n|^2 = 1.$$

For $a_n \neq 0$, write

$$p_n = |a_n|^2, \quad t_n = -\log |a_n| = \frac{1}{2} \log \frac{1}{p_n}.$$

Each coordinate of x in A_n has magnitude $|a_n| m_n^{-1/2}$, hence its negative logarithm is $s_n + t_n \geq 1$. Both arguments therefore lie in the power branch of ϕ_r , and disjointness gives the exact identity

$$\|(\mathcal{K}_{\phi_r} - L)x\|_2^2 = \sum_{n: p_n > 0} p_n ((s_n + t_n)^r - s_n^r)^2. \quad (1)$$

Split the indices into

$$I = \{n : t_n \leq s_n\}, \quad J = \{n : t_n > s_n\}.$$

For $n \in I$, concavity and the mean value theorem yield

$$(s_n + t_n)^r - s_n^r \leq r t_n s_n^{r-1}.$$

There is an absolute C such that

$$p \log^2(1/p) \leq C\sqrt{p} \quad (0 < p \leq 1).$$

Since $t_n = \frac{1}{2} \log(1/p_n)$, Cauchy–Schwarz gives

$$\begin{aligned} \sum_{n \in I} p_n ((s_n + t_n)^r - s_n^r)^2 &\leq r^2 \sum_n p_n t_n^2 s_n^{-\beta} \\ &\leq C_r \sum_n s_n^{-\beta} \sqrt{p_n} \\ &\leq C_r \left(\sum_n s_n^{-2\beta} \right)^{1/2} \left(\sum_n p_n \right)^{1/2} < \infty. \end{aligned}$$

The final series converges because $s_n \geq n^a$ and $2a\beta > 1$.

For $n \in J$, subadditivity of $t \mapsto t^r$ gives

$$(s_n + t_n)^r - s_n^r \leq t_n^r.$$

Also $p_n = e^{-2t_n}$. The function $e^{-2t}t^{2r}$ decreases for $t \geq r$; as $t_n > s_n \geq 1 > r$, we obtain

$$p_n t_n^{2r} = e^{-2t_n} t_n^{2r} \leq e^{-2s_n} s_n^{2r}.$$

Consequently,

$$\sum_{n \in J} p_n ((s_n + t_n)^r - s_n^r)^2 \leq \sum_n e^{-2s_n} s_n^{2r} < \infty.$$

Both bounds are independent of the normalized vector x . Equation (1) therefore proves

$$\|\mathcal{K}_{\phi_r}(x) - Lx\|_2 \leq C_r \|x\|_2 \quad (x \in Y),$$

so $\mathcal{K}_{\phi_r}|_Y$ is trivial.

For completeness, let $q : \ell_2 \oplus_{\mathcal{K}_{\phi_r}} \ell_2 \rightarrow \ell_2$ be the canonical quotient and define $T : Y \rightarrow \ell_2(\phi_r)$ by $T(x) = (Lx, x)$. Then $qT = \text{id}_Y$, while

$$\|T(x)\| = \|Lx - \mathcal{K}_{\phi_r}(x)\|_2 + \|x\|_2 \asymp \|x\|_2.$$

Thus $T(Y)$ is closed and $q|_{T(Y)}$ is an isomorphism onto Y . Hence q is not strictly singular. $\square$

Remark 1. The endpoint $r = 1$ is excluded for a structural reason: then $\beta = 0$, the summability mechanism disappears, and the classical Kalton–Peck centralizer is singular, as recalled in the source.

5 Verification and scope

The accompanying `verification.md` separately audits the block formula, both summability regions, the role of the ambient-valued linear map, and the passage from triviality to failure of strict singularity. No computational experiment is used as proof.

The result answers exactly the source’s question for every $0 < r < 1$. The block subspace depends on r , and no uniform subspace over the whole interval is claimed.

6 Bounded novelty check and review recommendation

On 19 July 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1410.5505, $\ell_2(\phi_r)$, sublinear Kalton–Peck centralizers, and the exact singularity phrase. Exact-phrase and title/author web searches were also run. The most relevant later primary result located, arXiv:2408.07827, concerns singular spaces built from bi-Lipschitz functions; it does not answer the sublinear t^r question. No matching answer was found. This bounded search does not establish originality.

Human-review recommendation. Treat this as a high-priority candidate full negative answer. The decisive checks are the convention allowing $L : Y \rightarrow \mathbb{C}^{\mathbb{N}}$ in triviality, the exponent $a > 1/(4(1 - r))$, and the explicit lifted copy $T(Y)$ in the twisted sum.

References

- [1] J. M. F. Castillo, V. Ferenczi, and M. González, *Singular twisted sums generated by complex interpolation*, arXiv:1410.5505v1 (2014), <https://arxiv.org/abs/1410.5505>.